INHERITANCE GENETICS

# Module map

- Module 09: Inheritance Genetics
- Connected lab: lab-09_inheritance-genetics.md
- Use this deck as the visual path through the module's evidence and retrieval work.

## Module 09

Inheritance Genetics

1. Alleles and genotype
2. Mendelian patterns
3. Punnett-square prediction
4. Beyond simple dominance
5. Pedigrees and probability

**Lab evidence**

lab-09_inheritance-genetics.md

INHERITANCE GENETICS

# Learning objectives

- Distinguish gene, allele, genotype, and phenotype.
- Use segregation to explain monohybrid inheritance.
- Build and interpret Punnett-square probabilities.
- Recognize inheritance patterns beyond complete dominance.
- Use pedigree evidence to reason about traits.

**OBJECTIVE 1**
Distinguish gene, allele, genotype, and phenotype.

**OBJECTIVE 2**
Use segregation to explain monohybrid inheritance.

**OBJECTIVE 3**
Build and interpret Punnett-square probabilities.

**OBJECTIVE 4**
Recognize inheritance patterns beyond complete dominance.

**OBJECTIVE 5**
Use pedigree evidence to reason about traits.

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# Topic sequence

- Alleles and genotype: Alleles are gene variants that contribute to genotype and phenotype.
- Mendelian patterns: Mendelian inheritance predicts patterns when alleles segregate into gametes.
- Punnett-square prediction: Punnett squares model probability, not guaranteed outcomes.
- Beyond simple dominance: Incomplete dominance, codominance, polygenic traits, and environment complicate simple patterns.
- Pedigrees and probability: Pedigrees use family evidence to infer inheritance patterns.

01
**Alleles and genotype**
Alleles are gene variants that contribute to genotype and phenotype.

02
**Mendelian patterns**
Mendelian inheritance predicts patterns when alleles segregate into gametes.

03
**Punnett-square prediction**
Punnett squares model probability, not guaranteed outcomes.

04
**Beyond simple dominance**
Incomplete dominance, codominance, polygenic traits, and environment complicate simple patterns.

05
**Pedigrees and probability**
Pedigrees use family evidence to infer inheritance patterns.

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# Module 09: Inheritance Genetics Evidence Map

- Module focus: Inheritance Genetics
- Central claim: Inheritance Genetics explains allele segregation with probability, phenotype prediction, and pedigree evidence.
- Key nodes to connect: Alleles and genotype, Mendelian patterns, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-09_inheritance-genetics.md supplies an evidence surface for the map.

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# Module 09: Inheritance Genetics Reasoning Sequence

- Module focus: Inheritance Genetics
- Trace the model: Alleles and genotype -> Mendelian patterns -> Punnett-square prediction -> Beyond simple dominance
- Inputs become outputs: Alleles and genotype, Mendelian patterns -> Distinguish gene, allele, genotype, and phenotype., Use segregation to explain monohybrid inheritance.
- Feedback check: lab-09_inheritance-genetics.md evidence checks whether students can explain punnett-square prediction.
- Lab connection: lab-09_inheritance-genetics.md gives students a process to observe or test.

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# Terms as evidence handles

- Gene: DNA sequence contributing to a trait or functional product.
- Allele: Variant form of a gene.
- Genotype: Allele combination of an organism.
- Phenotype: Observable trait or characteristic.
- Dominant: Allele effect visible with one copy in simple dominance.
- Recessive: Allele effect masked by dominant allele in heterozygotes.

**GENE**
DNA sequence contributing to a trait or functional product.

**ALLELE**
Variant form of a gene.

**GENOTYPE**
Allele combination of an organism.

**PHENOTYPE**
Observable trait or characteristic.

**DOMINANT**
Allele effect visible with one copy in simple dominance.

**RECESSIVE**
Allele effect masked by dominant allele in heterozygotes.

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# Lab connection

- Lab file: lab-09_inheritance-genetics.md
- Lab evidence should test or illustrate: Punnett-square prediction
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

**QUESTION**
Alleles and genotype

**EVIDENCE**
lab-09_inheritance-genetics.md

**CLAIM**
Distinguish gene, allele, genotype, and phenotype.

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# Contrast check

- Do not stop at: Alleles are gene variants that contribute to genotype and phenotype.
- Stronger explanation: Pedigrees use family evidence to infer inheritance patterns.
- The goal is to move from a named idea to a testable biological explanation.

**SURFACE**

Alleles are gene variants that contribute to genotype and phenotype.

**DEEPER**

Pedigrees use family evidence to infer inheritance patterns.

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# Module 09: Inheritance Genetics Retrieval and Lab Check

- Module focus: Inheritance Genetics
- Retrieval prompt: How are genes and alleles related?
- Answer check: Distinguish gene, allele, genotype, and phenotype.
- Required terms: Gene, Allele, Genotype
- Lab connection: lab-09_inheritance-genetics.md; lab-09_inheritance-genetics.md supplies evidence for crosses, pedigrees, and probability patterns in inheritance.

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# Practice quiz bridge

- 1. An organism's genotype is its:
- 2. Two heterozygotes are crossed (Aa x Aa). What fraction of offspring are expected to show the recessive trait?
- 3. A heterozygote (Aa) shows the dominant trait because:
- 4. What is a pedigree used for?

QUESTION 1

**Answer first**

An organism's genotype is its:

QUESTION 2

**Answer first**

Two heterozygotes are crossed (Aa x Aa). What fraction of offspring are expected to show the recessive trait?

QUESTION 3

**Answer first**

A heterozygote (Aa) shows the dominant trait because:

QUESTION 4

**Answer first**

What is a pedigree used for?

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# Synthesis and exit ticket

- Exit claim: explain alleles and genotype using evidence from lab-09_inheritance-genetics.md.
- Must include: Gene, Allele, Genotype.
- Revision prompt: what evidence would change your explanation?

**CLAIM**
Alleles and genotype

**EVIDENCE**
lab-09_inheritance-genetics.md

**REVISION**
What would change your mind?